

LUKE OFFSHORE BUOY 42040, USA

WMO: 995000

Lat: 29.208N

Lon: 88.226W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (W06)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.8	46.9	30.0	24.1	49.1	33.0	27.5	50.7	36.1	53.8	34.1	56.1	20.6	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	87.5	78.6	86.9	78.3	86.3	78.0	81.0	85.7	80.2	85.2	79.4	84.8	7.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.6	153.7	84.7	78.4	147.5	84.2	77.7	144.0	83.9	44.5	85.7	43.6	85.4	42.8	85.0	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.1	29.2	24.8	38.9	89.4	3.7	1.9	36.3	90.7	34.1	91.8	32.1	92.9	29.4	94.2		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.8	61.0	61.6	66.2	70.3	75.9	81.4	82.9	83.4	81.2	75.5	68.8	64.4
	DBStd	9.31	7.33	6.06	5.34	3.90	3.41	2.33	2.08	2.28	2.12	4.24	5.94	6.77
	HDD50	16	11	3	0	0	0	0	0	0	0	0	0	2
	HDD65	477	163	120	51	9	0	0	0	0	0	3	35	96
	CDD50	8330	350	328	503	608	804	943	1019	1036	936	791	563	449
	CDD65	3317	38	25	89	168	339	493	554	571	486	328	148	78
	CDH74	29772	1	0	12	176	2052	5532	6887	7286	5371	2057	341	57
	CDH80	8474	0	0	0	0	67	1487	2496	2910	1380	134	0	0
Wind	WSAvg	13.3	16.2	15.3	14.0	14.0	11.7	10.3	9.7	10.1	12.4	14.2	15.8	16.2
Precipitation	PrecAvg	57.8	4.7	3.8	4.8	4.8	3.9	5.1	6.5	6.5	5.9	4.1	3.5	4.2
	PrecMax	85.0	14.7	8.4	13.1	18.2	10.3	15.9	10.3	12.3	15.1	11.7	14.4	10.7
	PrecMin	35.2	0.1	0.7	0.6	0.5	0.4	1.1	3.0	2.5	1.8	0.7	0.4	1.1
	PrecStd	10.4	3.3	2.0	2.7	3.5	2.5	3.0	2.0	2.6	2.9	2.8	2.8	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.2	73.2	75.3	78.3	82.6	87.5	88.4	89.3	86.3	83.3	79.1	76.3
		MCWB	72.9	71.8	70.9	74.9	75.4	77.9	78.8	79.1	76.6	76.8	75.3	73.3
	2%	DB	73.2	71.8	74.3	77.1	81.6	86.4	87.2	88.0	85.4	82.4	78.0	75.6
		MCWB	71.5	69.7	70.4	73.6	75.4	77.6	78.5	78.8	76.5	76.6	73.3	72.1
	5%	DB	72.2	70.6	73.4	76.0	80.7	85.5	86.4	87.2	84.6	81.8	77.0	74.7
		MCWB	69.7	68.1	69.9	72.5	74.9	77.4	78.1	78.6	76.5	76.4	71.8	71.2
	10%	DB	71.0	69.3	72.6	75.1	80.0	84.7	85.9	86.6	84.2	80.5	76.1	73.5
		MCWB	67.8	66.6	69.2	71.5	74.6	76.9	77.8	78.3	76.4	75.0	70.3	70.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	72.5	72.9	76.4	78.3	80.6	81.4	82.5	80.4	81.2	77.3	74.4
		MCDB	73.6	72.8	74.0	77.7	80.6	84.0	86.3	87.2	83.1	81.5	78.6	75.4
	2%	WB	72.5	70.6	72.0	75.1	77.4	79.5	80.4	81.4	79.1	79.2	75.3	73.4
		MCDB	73.1	71.3	73.3	76.9	80.0	84.2	85.8	86.8	83.2	81.2	77.6	74.7
	5%	WB	70.4	68.8	71.1	73.8	76.7	78.7	79.7	80.4	78.4	78.1	73.1	72.5
		MCDB	71.8	69.9	72.8	75.5	79.6	83.8	85.5	86.1	83.1	80.8	76.4	74.3
	10%	WB	68.4	67.5	70.1	72.9	75.8	78.2	79.1	79.6	77.9	76.4	71.2	70.6
		MCDB	70.7	69.2	72.1	74.9	79.1	83.5	85.1	85.6	83.0	80.1	75.5	73.3
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.9	4.9	4.1	3.2	3.8	4.0	4.2	3.5	3.9	5.8	6.7
		MCDBR	5.7	7.0	4.2	4.1	3.4	4.2	4.1	4.5	3.7	4.1	5.4	6.7
	5% WB	MCWBR	7.3	8.8	5.0	4.9	3.4	3.9	3.6	3.9	4.1	5.6	6.9	8.5
		MCWBR	6.5	7.7	4.5	3.8	3.3	4.3	3.9	4.5	3.8	4.9	6.6	7.2
Clear-Sky Solar Irradiance	taub		0.353	0.367	0.379	0.412	0.436	0.453	0.479	0.477	0.439	0.395	0.366	0.357
		taud	2.524	2.493	2.452	2.351	2.326	2.328	2.274	2.287	2.352	2.468	2.491	2.523
	Ebn at Noon		280	285	288	281	273	266	259	259	266	272	273	273
		Edh at Noon	28	32	35	40	41	41	43	42	38	32	29	27
All-Sky Solar Radiation	RadAvg		988	1214	1623	1946	2189	2070	1912	1773	1666	1463	1126	888
	RadStd		85	161	136	181	108	156	123	140	104	122	110	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+238	+153

Nomenclature: See separate page